Parameter-efficient Fine-tuning with Orthogonal Fine-Tuning (OFT)

Mini-Project Report

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1. Introduction

Full fine-tuning of Large Language Models (LLMs) is computationally expensive. Parameter-Efficient Fine-Tuning (PEFT) methods address this by modifying only a small subset of parameters. **OFT** (Qiu et al., NeurIPS 2023) constrains weight updates to orthogonal transformations: $W' = R \cdot W$, where $R^T R = I$. Unlike LoRA's additive low-rank updates ($W' = W + BA$), OFT preserves the **hyperspherical energy** of pretrained weights, maintaining angular relationships between neurons. R is parameterized as a block-diagonal structure with r blocks for parameter efficiency.

In this project, we fine-tune **Qwen2.5-1.5B** (1.54B parameters) with OFT for **binary sentiment analysis** on SST-2, formulated as generative classification (the model generates "positive" or "negative").

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2. Experimental Setup

2.1 Configuration

Hyperparameter	Value
Pretrained Model	Qwen/Qwen2.5-1.5B
OFT Rank (r)	8
Target Modules	q_proj, v_proj
Learning Rate	1e-4
Batch Size	8
Epochs	3
LR Scheduler	Cosine with warmup (10%)
Optimizer	AdamW (weight_decay=0.01)
Precision	BFloat16
Max Sequence Length	128
Training Samples	5,000 (subsampled from 67,349)
Evaluation Samples	200 (from SST-2 validation set of 872)

OFT was applied to **q_proj** and **v_proj** in all attention blocks. Trainable parameters: **8,214,528 (0.53%** of 1.55B total). Hardware: NVIDIA RTX 5090 (32GB). Training time: **~5 min**.

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3. Results

3.1 Quantitative Results

Model	Accuracy
Qwen2.5-1.5B (base, zero-shot)	67.00%
Qwen2.5-1.5B + OFT (fine-tuned)	**96.00%**
Improvement	**+29.00%**

The OFT-fine-tuned model achieves a **96% accuracy** on the SST-2 evaluation set, representing a **+29 percentage point improvement** over the base model's zero-shot performance of 67%.

Figure 1: (Left) Accuracy before/after OFT. (Right) Parameter efficiency — only 0.53% trainable.

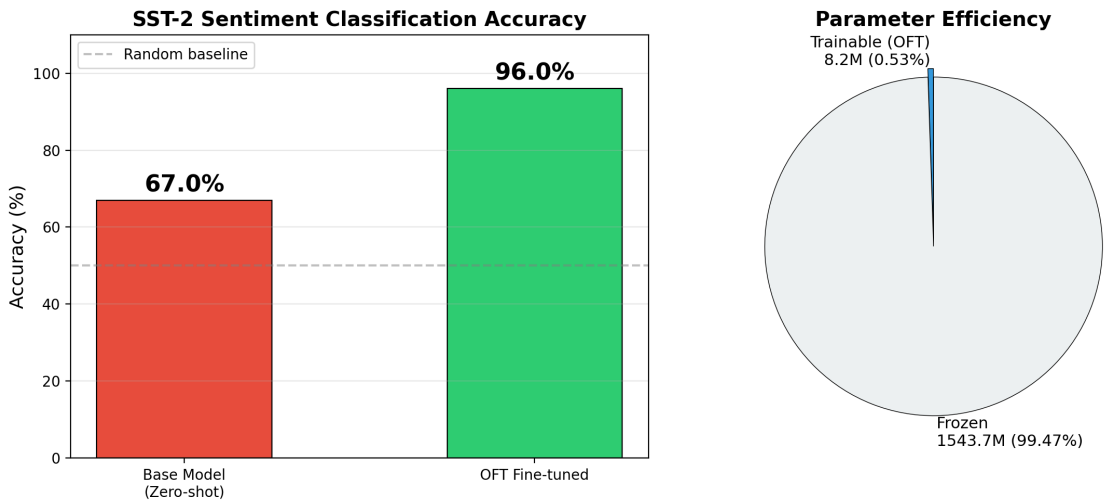

Figure 1: Accuracy Comparison and Parameter Efficiency

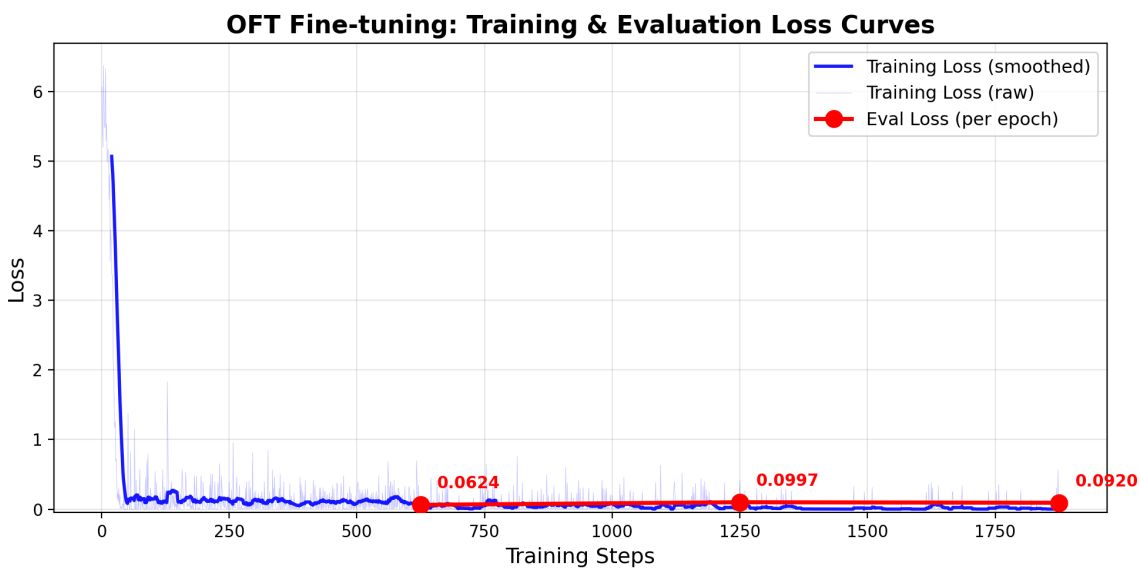

Figure 2: Training Loss Curve

3.2 Training Loss Curve

Figure 2: Training and eval loss over 3 epochs (1,875 steps).

Training loss drops rapidly (Epoch 1 avg: 0.30 → Epoch 3 avg: 0.015). Eval loss shows slight increase after Epoch 1 (0.062 → 0.092), suggesting mild overfitting, though generation accuracy continues to improve.

3.3 Qualitative Examples

Review (truncated)	GT	Base	OFT
"it's a charming and often affecting journey"	pos	pos ✓	pos ✓
"it takes a strange kind of laziness to waste the talents…"	neg	✗ (non-standard)	neg ✓
"we root for (clara and paul), even like them…"	pos	pos ✓	neg ✗
"a gorgeous, high-spirited musical from india…"	pos	pos ✓	pos ✓
"unflinchingly bleak and desperate"	neg	neg ✓	neg ✓
"in exactly 89 minutes, most of which passed as slowly…"	neg	neg ✓	neg ✓

Table 2: Qualitative examples. The base model occasionally produces non-standard outputs; OFT consistently generates correct sentiment labels (192/200 correct).

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4. Discussion and Conclusion

OFT demonstrates remarkable parameter efficiency: only **0.53%** of parameters are trainable, training completes in **~5 minutes**, and achieves **96% accuracy**. Compared to LoRA's additive updates, OFT's orthogonal constraint preserves hyperspherical energy (angular relationships between neurons), provides natural regularization against catastrophic forgetting, and may capture different adaptation patterns through multiplicative updates.

Limitations: Eval loss slightly increases after Epoch 1, suggesting overfitting on the 5K subsample; using the full 67K training set or early stopping could help. Only binary classification was tested.

Conclusion: By fine-tuning only 0.53% of Qwen2.5-1.5B's parameters with OFT, we achieved 96% accuracy on SST-2 — a +29% improvement over zero-shot — in under 5 minutes, confirming OFT's practical value as a PEFT technique.

References

1. Qiu, Z., Liu, W., Feng, H., et al. "Controlling Text-to-Image Diffusion by Orthogonal Finetuning." *NeurIPS 2023*.
2. Liu, B., et al. "Parameter-Efficient Orthogonal Finetuning via Butterfly Factorization." *ICLR 2024*.
3. Hu, E. J., et al. "LoRA: Low-Rank Adaptation of Large Language Models." *ICLR 2022*.
4. HuggingFace PEFT: https://github.com/huggingface/peft
5. Qwen2.5: https://huggingface.co/Qwen/Qwen2.5-1.5B
6. SST-2 Dataset (GLUE Benchmark): https://gluebenchmark.com/